

# Finite-Size Majorana Splitting in the Kitaev Chain: Why do the “spikes” in the finite-size spectrum worsen with larger $L$ ?

Seminar notes

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## 1 Setup

We study the Kitaev chain at  $t = \Delta = 1$  with open boundary conditions (OBC). The BdG Hamiltonian is the real-symmetric matrix

$$M = \begin{pmatrix} h & d \\ -d & -h \end{pmatrix}, \quad h_{ij} = -\mu\delta_{ij} - t(\delta_{i,j+1} + \delta_{i+1,j}), \quad d_{ij} = \Delta(\delta_{i,j+1} - \delta_{i+1,j}). \quad (1)$$

Particle-hole symmetry (PHS) guarantees eigenvalues come in exact  $\pm E$  pairs. Plot 4 tracks the *six smallest positive eigenvalues* as  $\mu$  varies.

## 2 The ideal edge-mode energy

### 2.1 Zero-energy condition in the complex BZ

To find the edge-mode energy, we look for a zero-energy solution that decays into the bulk. Set  $E = 0$  and try an Ansatz  $\psi_j \sim \lambda^j$ . Substituting into the bulk BdG equations gives the characteristic equation

$$(-\mu - t(\lambda + \lambda^{-1}))^2 + \Delta^2(\lambda - \lambda^{-1})^2 = 0. \quad (2)$$

For  $t = \Delta$  this factors to

$$(-\mu - t(\lambda + \lambda^{-1}) + \Delta(\lambda - \lambda^{-1}))(-\mu - t(\lambda + \lambda^{-1}) - \Delta(\lambda - \lambda^{-1})) = 0. \quad (3)$$

The two branches give the solutions

$$\lambda_A = \frac{\mu}{2t}, \quad \lambda_B = \frac{2t}{\mu}. \quad (4)$$

For the left edge mode we need  $|\lambda| < 1$ , so we take  $\lambda = \mu/(2t)$  (valid for  $|\mu| < 2t$ , i.e. inside the topological phase). The wavefunction decays *purely exponentially*:

$$\psi_j^L \sim \left(\frac{|\mu|}{2t}\right)^{j-1}, \quad \xi = \frac{1}{|\ln |\mu/2t||} \quad (\text{coherence length}). \quad (5)$$

Crucially, for  $t = \Delta$  the solution has **no oscillatory factor**  $e^{ik_0j}$ . This is special to the symmetric point; for  $t \neq \Delta$  the roots can be complex, introducing oscillations.

## 2.2 Finite-size hybridization splitting

A chain of length  $L$  has both a left edge mode  $\psi_j^L \sim \lambda^{j-1}$  and a right edge mode  $\psi_j^R \sim \lambda^{L-j}$ . Their overlap determines the splitting of the zero-energy pair:

$$\delta E \approx 2t \frac{\langle \psi^L | V | \psi^R \rangle}{\|\psi^L\| \|\psi^R\|} \sim 2t \lambda^L = 2t \left( \frac{|\mu|}{2t} \right)^L. \quad (6)$$

This is the energy of the near-zero mode. For  $t = \Delta = 1$ :

$$E_0(\mu, L) \approx 2 \left( \frac{|\mu|}{2} \right)^L. \quad (7)$$

Predictions from (7):

- At the sweet spot  $\mu = 0$ :  $E_0 = 0$  *exactly*, for any  $L$ .
- At the phase boundary  $|\mu| = 2t$ :  $E_0 \rightarrow 2$  (the bulk gap), for any  $L$ .
- In the interior:  $E_0$  decreases *exponentially* with  $L$ . A larger chain has a better-protected zero mode.

So (7) predicts that larger  $L$  should give *smaller* spikes, not larger ones.

## 3 Why the spikes increase and widen with $L$ : a numerical artifact

### 3.1 The floating-point threshold problem

The observed behaviour (spikes worsen as  $L$  increases) is *not* a physical effect — it is a numerical artefact arising from how we define “positive spectrum”.

In the simulation, positive eigenvalues are selected with the filter `evals[evals >= 0]`. For large  $L$ , the ideal splitting  $E_0 = 2(|\mu|/2)^L$  falls below machine precision ( $\approx 10^{-15}$ ) over a wide range of  $\mu$ :

$$E_0 < \varepsilon_{\text{mach}} \iff \left( \frac{|\mu|}{2} \right)^L < \frac{\varepsilon_{\text{mach}}}{2} \iff |\mu| < 2\varepsilon_{\text{mach}}^{1/L}. \quad (8)$$

For  $\varepsilon_{\text{mach}} = 5 \times 10^{-16}$ :

$$L = 30 : \quad |\mu|_{\text{thresh}} \approx 0.62, \quad (9)$$

$$L = 100 : \quad |\mu|_{\text{thresh}} \approx 1.41. \quad (10)$$

For  $|\mu| < |\mu|_{\text{thresh}}$ , the near-zero eigenvalue is indistinguishable from zero at double precision. Due to rounding, it can be stored as a tiny *negative* number ( $\sim -10^{-16}$ ). The filter `evals >= 0` then *excludes* this eigenvalue, and the *lowest bulk mode* (at finite energy  $\sim E_{\text{gap}}$ ) takes its place as the “lowest positive eigenvalue”. This appears as a spike.

### 3.2 Width and height of the artefact

- **Width:** the spike region is  $|\mu| < 2\varepsilon_{\text{mach}}^{1/L}$ , which *grows* with  $L$  (for  $L = 100$  it covers most of the topological phase, explaining the “widened” spikes).
- **Height:** the spike jumps to the lowest bulk energy, which is  $\sim E_{\text{gap}}(\mu)/(L\text{-dependent level spacing})$ . With larger  $L$  the bulk level spacing is smaller, so more bulk levels fall inside the window and the spikes look denser.

## 4 The fix: use quasiparticle energies $|E|$

The physical quasiparticle spectrum consists of the *absolute values* of the BdG eigenvalues. The near-zero mode has  $|E| \approx 0$  regardless of the sign of the floating-point result:

$$\text{correct : } \{E_n\}_{n=1}^L = \text{one copy from each adjacent pair in sort}(|e_i|). \quad (11)$$

In code, replace

```
return evals[evals >= 0]          # WRONG: sign-sensitive
```

with

```
return np.sort(np.abs(evals))[:,2] # correct: |E| plus pair deduplication
```

The absolute value is what protects the near-zero Majorana mode from a floating-point sign flip. The stride is what removes the particle-hole double counting: each physical quasiparticle energy appears twice as  $\pm E$  before taking the absolute value.

## 5 Summary

| Observation                        | Physical or artefact? | Cause                                                             |
|------------------------------------|-----------------------|-------------------------------------------------------------------|
| Spikes near $ \mu  \rightarrow 2t$ | <b>Physical</b>       | $\xi \rightarrow \infty$ , hybridisation is strong                |
| Spikes in interior for small $L$   | Mostly artefact       | Float precision, level crossings                                  |
| Spikes widen/worsen for large $L$  | <b>Artefact</b>       | Threshold $ \mu _{\text{thresh}} \propto \varepsilon^{1/L}$ grows |